Constructing a Structural Model of Algebra Task Difficulty With Implications for Design of Digital Teaching

Joseph Taylor
MA32050
Supervised by Waleed A Ali

Abstract

This study develops an explanatory Rasch model for analysing the structural difficulty of tasks used in secondary school (or Key Stage 3 (KS3)) algebra. While standard *Item Response Theory* (IRT) provides robust estimates of task difficulty on a logit scale, it treats difficulty as a single latent parameter without explaining which mathematical features drive it. Conversely, *Cognitive Load Theory* and research on mathematical representations provide theoretical accounts of why algebra tasks vary in cognitive demand, but lack a formal measurement structure for measuring these effects. To address this gap, this project proposes a structured explanatory Rasch model in which task difficulty is expressed as a function of observable algebraic features such as operational complexity and representational transitions. These features act as proxies for element interactivity and intrinsic cognitive load.

The framework is illustrated through worked examples and evaluated via simulation using response data generated under realistic conditions. The results demonstrate that the model can decompose real algebraic difficulty and quantify how changes in task structure (e.g., symbolic variety) and task design (e.g., hover-over hints) are predicted to alter effective cognitive demand. This provides a mathematics-specific approach for evaluating the effect of design choices on the difficulty of digital algebraic tasks, of use prior to large-scale empirical testing of new instructions. This project maintains a clear distinction between task difficulty and learning outcomes; reductions in difficulty do not necessarily imply improved learning.

1 Introduction

1.1 Background and motivation

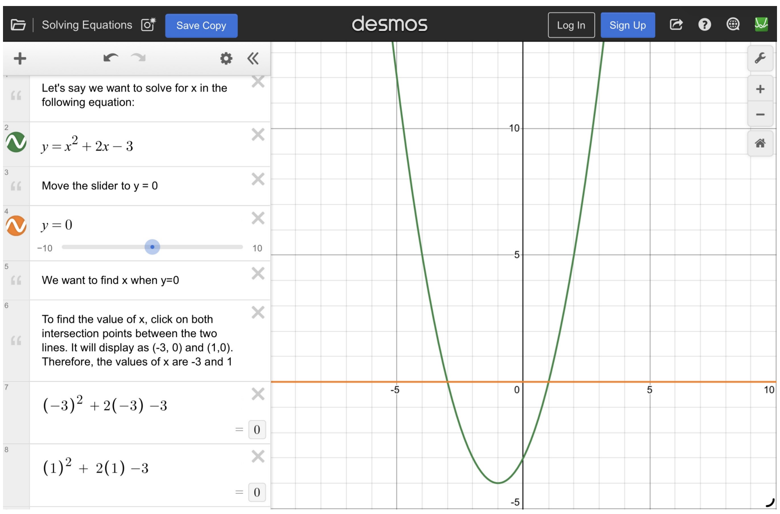
Digital teaching platforms increasingly incorporate interactive features such as dynamic graphing tools, hover-over hints, and click-through explanations. These are often assumed to reduce cognitive load and support algebraic reasoning (Sweller, 1988; Mayer, 2009). However, it remains unclear whether such features genuinely simplify tasks or merely change their nature, and whether any observed performance gains stem from the improved pedagogical effectiveness of the task design itself, or other factors such as student familiarity with interfaces.

Current methods for evaluating instructional design rely heavily on subjective measures which capture learners' perceived mental effort and difficulty (Paas, 1992; Hart and Staveland, 1988). They are applied after students have completed tasks, making them time-consuming and of limited value for proactive task design. Given the limited teaching time available in schools, evaluating the effectiveness of teaching and assessment approaches cannot rely solely on retrospective trial-and-error methods.

To improve digital algebra teaching, we must first understand what makes a task difficult. Consider the same quadratic task presented in static symbolic form versus an interactive Desmos version with sliders and dynamic graphs (Figure 1.1): What makes one task easier than the other?

$$x^2 + 2x - 3 = 0$$

Symbolic version



Desmos interactive version

Figure 1.1: Comparison of symbolic and digital presentation of the same quadratic task.

Although the interactive version appears more accessible, additional scaffolding and interactivity do not necessarily reduce cognitive demand and may alter the nature of the task itself. A more accessible path to an answer does not guarantee improved conceptual understanding. It is therefore not sufficient to assume that such features reduce difficulty and support learning without systematic evaluation. This motivates the need for a model capable of distinguishing structural complexity from design effects when evaluating whether digital formats meaningfully change task difficulty (Bond and Fox, 2015).

1.2 Overview

Algebra is widely recognised as one of the most challenging areas of secondary mathematics. Tasks require students to coordinate multiple linked elements, follow structured procedures and maintain focus. Tasks vary in how many operations they contain, how symbols interact and how often students must move between different representational forms such as equations and graphs. These structural features impose significant demands on working memory, as described by Cognitive Load Theory.

Digital tools such as GeoGebra and Desmos are now commonly used in algebra teaching. They can dynamically link representations and provide scaffolding, such as hover-over hints or interactive graphing tools, to change the cognitive demands of the task while the underlying maths remains the same. But empirical findings are mixed: some studies suggest improved conceptual understanding through visualisation, while others highlight risks of distraction, superficial processing, or increased extraneous load. Most existing research evaluates digital instruction through overall attainment outcomes, or through the student's perception of difficulty, rather than analysing how the structural and representational design features of algebra tasks influence mental effort by contributing to task difficulty.

In educational measurement, models such as the Rasch model (Rasch, 1960) estimate task difficulty from student performance data. Explanatory extensions to Item Response Theory (IRT) allow difficulty to be linked to observable task features, but algebraic structure and representational transitions are rarely formalised within these models. At the same time, Cognitive Load Theory explains why managing multiple interacting elements increases mental effort, yet it does not provide a mathematical way to measure symbolic structure. There is therefore a gap between (i) statistical models of performance, (ii) cognitive theories of task demand, and (iii) research on digital algebra instruction.

This project addresses the gap by developing and analysing a structured explanatory Rasch model in which task difficulty is expressed in terms of observable structural and representational features of algebra tasks.

1.3 Research questions (RQ)

- (RQ1): How does existing research describe the ways in which digital presentation may alter the cognitive demands of secondary school algebra tasks?
- (RQ2): How can the structural features of algebra tasks, such as symbolic complexity and representational transitions, be expressed within a mathematical model of task difficulty?
- (RQ3): How can such a model be used to compare algebra tasks presented in digital and non-digital formats?
- (RQ4): How could this model provide a basis for testing hypotheses about the effects of digital presentation on algebra performance?

1.4 Scope and approach

This project takes a mathematical modelling approach to understanding the difficulty of KS3 algebraic questions. A Rasch-style item response model is used to interpret task (item) difficulty as a measurable parameter. Cognitive load theory provides the explanation for why structural features, such as symbols, multiple linked steps, or representational features make tasks more demanding. These features are expressed using measurable indicators so that task complexity can be described clearly and consistently. This enables systematic comparison between tasks. By placing task difficulty and student ability on a common logit scale, the model allows differences in performance to be interpreted as differences in task design rather than underlying attainment (Rasch, 1960; Wright and Stone, 1979).

This project focuses on instructional task design to support learning, rather than on assessment tasks for measuring outcomes (De Boeck and Wilson, 2004; Embretson and Reise, 2000). The aim is not to predict individual student performance, but to clarify how features of task design contribute to mental effort.

The main contribution of this project is therefore analytical rather than experimental. It develops the model, examines its theoretical properties, and illustrates how it works using worked algebra examples. A small simulation study is included to check that the model behaves as expected under controlled conditions. The findings are then used to discuss implications for the design of algebra tasks in both digital and non-digital classroom settings.

2 Literature review

2.1 Literature search method

A structured literature search was conducted to identify quantitative and modelling-based research relevant to algebraic task difficulty, cognitive load, representation, and item response theory. Early scoping searches suggested that, although these areas are well developed individually, there is no unified mathematical model that decomposes algebra task difficulty into explicit structural and representational components within an item response framework. This observation shaped the direction of this study.

Searches were conducted using JSTOR and Google Scholar, alongside targeted searches in leading journals including Educational Studies in Mathematics, Journal for Research in Mathematics Education, Learning and Instruction, and Computers and Education. Foundational theoretical works were included regardless of date, while empirical studies were mainly drawn from 2000–2025. Search terms included combinations of: “task difficulty”, “algebra”, “representation”, “mathematics learning”, “task complexity”, “mathematics education”, “Rasch model”, “mathematics”, “item response theory”, “cognitive load”, “mathematics” and “element interactivity”.

Studies were included if they: used explicit quantitative or statistical modelling, examined algebraic tasks or symbolic representations, operationalised cognitive load or structural features in measurable terms, and were peer-reviewed and published in English.

The final selection included foundational work in item response theory (particularly within psychometric frameworks such as Rasch modelling and explanatory item response theory), cognitive load research on element interactivity, representation frameworks relevant to algebra, and quantitative studies of digital algebra environments. Together, these sources inform the structured model developed in Chapter 3.1.

2.2 Why task difficulty matters in education

Understanding task difficulty is central to effective mathematics teaching. The level of challenge influences engagement and conceptual understanding. Cognitive load theory argues that learning is constrained by working memory capacity, and that tasks that are too complex may overload learners and hinder understanding (Sweller, 1988; Sweller et al., 1998). Conversely, tasks that are too simple may fail to promote meaningful learning; effective instruction therefore depends on calibrating task demand appropriately (Sweller et al., 1998; Kalyuga et al., 2003). In secondary algebra, teachers regularly adjust examples, representations, and scaffolds to modify task difficulty. However, such adjustments are often guided by experience rather than by an explicit structural account of what makes one algebra task more demanding than another.

Existing theory suggests that task difficulty is influenced by (i) intrinsic structural features that place demands on working memory (cognitive load theory), (ii) the effort required to convert or coordinate between representations such as symbols and graphs (representation research), and (iii) presentation and scaffolding choices that can add avoidable load or support learning (instructional design research). However, these ideas have not been combined within a single model. This chapter reviews the theoretical and modelling perspectives on task difficulty and identifies the absence of a framework breaking down algebraic task difficulty into measurable components.

2.3 Algebraic difficulty: symbolic structure and representational forms

Research on the representation of mathematics provides a route to understanding task difficulty. In algebraic expressions, symbols represent operations and relationships in a condensed and structured way (Tall, 2004), which require manipulation during problem-solving. This supports the idea that symbolic structure itself should be analysed explicitly when modelling algebraic tasks.

Duval's theory of semiotic registers argues that mathematical understanding requires conversion between representational forms, such as symbols, graphs, and descriptions. These conversions are cognitively demanding and may increase task difficulty independently of symbolic manipulation. For example, interpreting a quadratic expression graphically introduces representational effort beyond the algebraic manipulation itself (Duval, 2006).

Ainsworth's DeFT framework (design, functions, tasks) and Mayer's multimedia principles show that multiple representations can either support or overload learning (Ainsworth, 2006; Mayer, 2001). While representations can complement each other, converting between them involves additional cognitive demands and can risk overloading working memory (Ainsworth, 2006). This is particularly relevant in digital algebra environments, where symbolic and graphical forms are presented together and linked dynamically (Ainsworth, 2006), increasing the demands of the task. Despite this, representational transitions are rarely incorporated directly into statistical models of task difficulty (De Boeck and Wilson, 2004; Duval, 2006).

2.4 Cognitive foundations of difficulty

Cognitive load theory contends that learning is constrained by the limits of working memory, and provides a framework for explaining why some algebra tasks are more difficult than others (Sweller, 1988; Sweller et al., 1998). It distinguishes between three types of cognitive load: intrinsic load, which arises from the inherent structure of the task; extraneous load, which is generated by the way information is presented; and germane load, which relates to the cognitive processes involved in constructing understanding (Sweller et al., 1998). This distinction is important for this study, as it separates structural sources of difficulty from those introduced through task design.

Intrinsic load, element interactivity and symbolic complexity

Intrinsic load increases when more elements of a task must be processed at the same time, known as *Element Interactivity* (Sweller, 1988). In algebra, greater symbolic complexity, and multiple operations increase intrinsic load – the cognitive demand placed on working memory — and therefore contribute to task difficulty (Sweller, 1988).

However, cognitive load is not directly measurable. Previous work has estimated cognitive load using ratings or performance measures (Paas, 1992; Paas et al., 2003), but these approaches treat load as a psychological concept rather than deriving it from task structure. In this project, observable structural features of algebra tasks are therefore used as proxies for intrinsic load, enabling these effects to be incorporated into a quantitative model of task difficulty.

Extraneous load, germane load and scaffolding

Scaffolding provides temporary support enabling task completion beyond independent capability (Wood et al., 1976). Within mathematics education, scaffolding comprises a set of design features that guide attention, reduce unnecessary complexity, and support problem-solving processes. From a cognitive perspective, scaffolding works by modifying extraneous cognitive load and supporting the management of both germane and intrinsic load (Sweller et al., 1998). It can alter the difficulty of a task without changing its underlying structure. For example, breaking a task into intermediate steps reduces the need to coordinate multiple elements simultaneously. However, poorly designed guidance may reduce effective cognitive processing, limiting deeper learning and transfer (Kirschner et al., 2006; Hofer et al., 2025).

In digital environments, scaffolding is often embedded directly within tasks through interactive features such as stepwise hints, dynamic linking of representations to assist with transition, and automated feedback (Ainsworth, 2006; Hillmayr et al., 2020). These features may reduce cognitive demand by structuring problem-solving processes or making relationships explicit, but may also introduce additional demands or encourage dependency if poorly designed. This reinforces the view that scaffolding can both reduce and increase effective task difficulty depending on its design.

Scaffolding is operationalised in this project using three dimensions: conceptual, procedural, and metacognitive support. Each is treated as an observable feature of task design that can be quantified and incorporated into a model of task difficulty, as described in Appendix A. This categorisation aligns closely with established distinctions in mathematics education, including Skemp (1976). Procedural scaffolding guides the steps required to complete a task, so has immediate effect. Conceptual scaffolding supports understanding of underlying mathematical relationships, and metacognitive scaffolding supports planning, monitoring, and evaluation of problem-solving processes.

Learner ability and expertise reversal

Kalyuga's expertise reversal effect further shows that the impact of task design depends on learner ability (Kalyuga et al., 2003): instructional supports which reduce cognitive load for novices may be unnecessary or even detrimental for more advanced learners. This indicates that effective difficulty depends not only on task structure but also on learner ability. This insight is relevant when considering digital algebra tools, as scaffolding that reduces effective complexity for beginners may alter difficulty differently depending on learner ability. This highlights the importance of modelling both task features and student ability within a common framework (Kalyuga et al., 2003; Sweller, 1988).

2.5 Digital algebra environments

Following the shift to online learning during the COVID-19 pandemic, there has been a rapid expansion of research into digital education, online mathematics instruction and scaffolding (see for example, Zuo et al. (2023); Malik et al. (2025) and Farmer (2024); Larico et al. (2026)). These review scaffolding implemented in various forms of online mathematics learning including GeoGebra and Desmos, building on earlier theoretical work.

Digital technologies are widely used in secondary algebra teaching. Research in technology and mathematics education examines how digital tools reshape algebraic activity and representation use (Hoyles and Lagrange, 2010). Digital applications allow symbolic expressions and graphical representations to be manipulated and linked in real time. This may reduce some representational barriers, but may also introduce new sources of difficulty if tasks are poorly designed (Hoyles and Lagrange, 2010; Mayer, 2001).

Empirical research findings are mixed. Some studies report improved conceptual understanding when digital tasks are carefully designed to help students connect equations, graphs, and other forms, while other studies highlight risks of superficial engagement or increased distraction (Hoyles and Lagrange, 2010; Zuo et al., 2023; Khalil et al., 2019; Sweller, 1988). A limitation is that many studies evaluate overall achievement outcomes without modelling how the structural or representational complexity of particular algebra tasks changes across digital and non-digital formats (De Boeck and Wilson, 2004; Duval, 2006).

Subjective measures of cognitive load in digital tasks

Current approaches to assessing the effects of instructional design on cognitive load and task difficulty rely heavily on subjective rating scales, such as the single-item 9-point cognitive load scale (Paas, 1992) and the NASA Task Load Index (NASA-TLX) (Hart and Staveland, 1988) which capture learners' perceived mental effort and difficulty, often supplemented by performance indicators and, less frequently, physiological measures.

Several studies have used Paas's scale to compare cognitive load when students solve systems of equations or explore functions using dynamic graphing software versus static worksheets. For example, Farmer (2024) found that students solving advanced algebra using Desmos reported significantly lower mental effort ratings compared to paper-based methods, attributing the reduction to Desmos's ability to dynamically link symbolic and graphical representations, making transitions easier. Similarly, Khalil et al. (2019) reported lower perceived cognitive load when students used GeoGebra to explore analytic geometry, particularly in tasks requiring coordination between algebraic and visual representations. Both of these studies were focussed on upper secondary school aged children and, importantly, involved assessment of perceived difficulty after task completion.

While these measures are commonly used to assess perceived difficulty and mental effort, they were not originally developed for mathematics or digital formats and are used retrospectively after learners have completed the task. This limits their usefulness for upfront design of instructional tasks and highlights the need for analytical tools, such as the structural difficulty model proposed in this study, that can predict cognitive demand at the task design stage.

2.6 Quantitative measurement of task difficulty

Rasch model

Many quantitative models in mathematics education use statistical measurement models, most notably the Rasch model (Rasch, 1960). In this model, the probability that a student answers a question correctly depends on the difference between the student's ability and the difficulty of the question. A key strength is that both ability and difficulty are placed on the same logit scale, allowing meaningful comparison of tasks (Rasch, 1960; Wright and Stone, 1979).

The task (item) difficulty parameter corresponds to a point on the latent ability (logit) scale at which a student whose ability equals that value has a 50% probability of success (Rasch, 1960; Wright and Stone, 1979; Embretson and Reise, 2000). This reflects the probabilistic nature of human performance; even well-matched tasks may be answered incorrectly due to variation in understanding, strategy, attention, or error.

A limitation is that the Rasch model treats difficulty as a single numerical value estimated from response data. Accordingly, two linear algebra questions may be estimated to have similar difficulty despite differing substantially in complexity. Applied work on Rasch measurement highlights the importance of careful interpretation of difficulty estimates and model fit in testing contexts (Wright and Masters, 1982), but the model does not in itself explain which features of a task contribute to difficulty (Wright and Masters, 1982; Wright and Stone, 1979).

Explanatory Item Response Theory (IRT)

While the Rasch model captures the overall difficulty of a task, it does not capture the sources of that difficulty or allow detailed comparison between different tasks (Rasch, 1960; Wright and Masters, 1982). Explanatory item response theory addresses this limitation by modelling task difficulty as a function of observable task features (De Boeck and Wilson, 2004). In particular the Linear Logistic Test Model (LLTM) allows item difficulty to be represented as a linear combination of specified features (Fischer, 1973). Embretson's work on explanatory IRT enables extension of this approach by linking cognitive and structural task features to estimated difficulty parameters (Embretson and Reise, 2000).

LLTM-based approaches enable task difficulty to be explained, but assume that it can be represented as a linear combination of specified features, which may oversimplify the complexity of cognitive processes involved. Despite this, algebra-specific breakdowns of task structure and difficulty remain limited in mathematics education research (De Boeck and Wilson, 2004; Embretson and Reise, 2000).

Alternative predictive models

Educational data mining and learning sciences research often uses probabilistic models to describe how student performance develops over time, for example Bayesian knowledge tracing (Corbett and Anderson, 1994) and learning curves (Koedinger et al., 2023). These models represent learning as transitions in an underlying knowledge state (Corbett and Anderson, 1994). However, they are not designed to explain task difficulty in terms of internal task structure. They therefore provide useful comparisons but do not address structural decomposition of algebraic task difficulty (Corbett and Anderson, 1994; De Boeck and Wilson, 2004).

The Rasch model assumes equal discrimination (a *One-Parameter Logistic Model*, or 1PL), meaning all tasks are treated as equally effective at distinguishing between higher-and lower-ability students. More complex models, such as the two-parameter logistic (2PL) model, allow discrimination to vary across tasks. However, this project focuses on modelling differences in task difficulty rather than discriminating by ability, and therefore adopts the Rasch formulation to avoid unnecessary complication.

Simulations in Rasch and LLTM frameworks

Simulation studies are commonly used in Rasch and LLTM frameworks to evaluate whether underlying model parameters can be reliably recovered from response data. Within explanatory IRT, simulation is used to test whether task features can be reliably estimated (Embretson, 1984; De Boeck and Wilson, 2004).

Previous studies have examined specific item-level effects. For example, Debeer and Janssen (2013) simulate response data to examine the recovery of item-position effects, demonstrating how an additional explanatory variable can be incorporated into the model and tested through repeated sampling. Similarly, studies such as Janssen, De Boeck, et al. (2004) and Kubinger (2005) use simulation to assess the validity of LLTM assumptions and to compare the performance of feature-based models with standard Rasch estimation. Typically, simulation involves (i) defining a data-generating model with known parameters, (ii) generating realistic response data according to the logistic probability function, and then (iii) assessing whether the estimation procedure can recover the underlying parameters.

This study adopts the same simulation-based approach, but applies it to a new context. By incorporating scaffolding as an item-level variable within a Rasch/LLTM framework, the simulation investigates whether its effect on task difficulty can be detected and reliably estimated.

2.7 Identified gap in research

There are consistent limitations across these studies: Statistical models measure how difficult a task is, but they do not explain its sources (Rasch, 1960; Embretson and Reise, 2000; De Boeck and Wilson, 2004); cognitive load theory explains why tasks become harder as element interactivity increases, but does not provide a measure for algebraic complexity (Sweller, 1988; Paas et al., 2003); representation frameworks explain that moving between equations, graphs, and words increases cognitive demand, but this is rarely built into models that measure difficulty (Duval, 2006; Ainsworth, 2006); research on digital-algebra environments evaluates outcomes without modelling how task structure changes across formats (Hoyles and Lagrange, 2010), despite digital scaffolding research expanding since 2020. There is therefore a need for a framework that combines structural, representational and design components within an item response model.

This project develops such a framework for use in mathematics education, by combining item response modelling, cognitive load theory, and representation research within a single model. The approach enables the difficulty of algebra tasks to be expressed in terms of observable task features and allows instructional design choices, including digital scaffolding, to be evaluated within a common measurement scale at the design stage. A simulation-based methodology is then used to evaluate whether the design effects can be reliably identified in practice under controlled conditions using the proposed model.

3 A model for algebra task difficulty

3.1 Mathematical framework

In accordance with the Rasch model, in this dissertation, "difficulty" is defined as the estimated difficulty parameter of a task within an item response model, represented on a common logit scale. It represents the level of ability required to achieve a given probability of success and is therefore a relational property between task and learner (Rasch, 1960; Wright and Stone, 1979). By contrast, cognitive load refers to the mental effort required during task performance (Sweller, 1988). Although distinct, these concepts are related; tasks that impose greater cognitive demand would be expected to exhibit higher estimated difficulty. The proposed model links structural and representational features of algebra tasks to task difficulty, treating these features as observable proxies for underlying cognitive demand (Sweller et al., 2011; Paas et al., 2003).

The explanatory model developed in this study is based on the Rasch (1PL) model and its LLTM extension (Fischer, 1973; De Boeck and Wilson, 2004). It therefore inherits these core assumptions:

- **Unidimensionality:** A single latent trait (KS3 algebra ability) accounts for the variation in student responses;
- **Local independence:** Given a student's ability level, responses to different tasks are statistically independent;
- **Parameter invariance:** The difficulty parameters (and the feature coefficients in the LLTM) do not depend on the particular sample of students, allowing generalisable inferences about task features;
- **Fixed discrimination:** All items are assumed to have the same discrimination parameter;
- **No pseudo-guessing:** There is no lower asymptote, which is reasonable for algebra tasks where guessing is not a major factor.

These assumptions enable strong measurement properties and allow task difficulty to be decomposed into structural and design-related components. Potential violations (e.g., multidimensionality due to shared contexts across tasks, or dependence introduced by digital scaffolding) are acknowledged as limitations in Chapter 7.

Theoretical framework

To apply the Rasch and LLTM models for this study, four complementary theoretical strands are used to provide justification for representing task difficulty in terms of observable features:

- *Item response theory* (Rasch model and explanatory IRT) provides the foundation for modelling task difficulty and ability on a common logit scale (Wright and Stone, 1979). Explanatory IRT, in particular LLTM (Fischer, 1973) justifies expressing difficulty as a function of observable task features rather than treating it as an unexplained parameter. Because the Rasch model separates ability and difficulty differences in performance cannot simply be attributed to stronger or weaker students, instead, they appear as changes in estimated task difficulty b_i arising from task features. A key property of the model is that task difficulty can be interpreted independently of the specific sample of students.

- *Cognitive load theory* (Sweller) explains why structural features of algebra tasks increase difficulty. Increased mental effort arises from greater "element interactivity", that is, the number of interacting elements that have to be processed simultaneously (Sweller, 1988). Algebraic tasks involving nested operations and interacting symbols require greater cognitive load, prompting the inclusion of the structural variables of symbolic complexity, operational count, and dependency depth within the model. These features are treated as observable indicators of the cognitive demand imposed by the task. Within the LLTM framework, they contribute additively to task difficulty through their associated coefficients.

- *Scaffolding research* (Wood et al., 1976; Skemp, 1976) explains how supportive features can reduce cognitive demand, providing theoretical support for inclusion of instructional design features that may reduce estimated task difficulty. In this study, scaffolding is represented through procedural, metacognitive, adaptive and conceptual features that structure the task, guide solution processes, or support coordination. We capture the presence of scaffolding rather than its pedagogical effectiveness, as we are not examining learning outcomes. Crucially, a reduction in estimated difficulty does not necessarily imply improved learning (Sweller et al., 2011; Kirschner et al., 2006; Bjork, 1994; Paas et al., 2003). This project therefore focuses on modelling task difficulty, rather than making direct claims about learning outcomes.

- *Representation frameworks* (Tall, Duval) explain how transitions between symbolic and graphical forms require additional mental effort and coordination. This motivates the inclusion of representational variables within the model, potentially contributing to task difficulty by increasing the coordination required during task performance.

Figure 3.1 illustrates the range of factors that may influence the observed difficulty of an algebra task. The proposed model focuses on the structural and design features of the task itself, while recognising that other external influences, such as prior attainment and digital familiarity, may also affect performance but are not explicitly modelled.

3.2 Development of the model

This section develops the model step by step. We begin with the standard Rasch model, then introduce observable features of algebraic tasks, and then construct a model in which task difficulty is explained in terms of these features.

Base item response model

Let X_{pi} denote the outcome of student p answering an algebra test task i:

$$X_{pi} = \begin{cases} 1 & \text{if student } k \text{ answers task } i \text{ correctly,} \\ 0 & \text{otherwise.} \end{cases}$$

The Rasch model (Rasch, 1960) expresses the probability of a correct response as:

$$P(X_{pi} = 1) = \sigma(\theta_p - b_i)$$

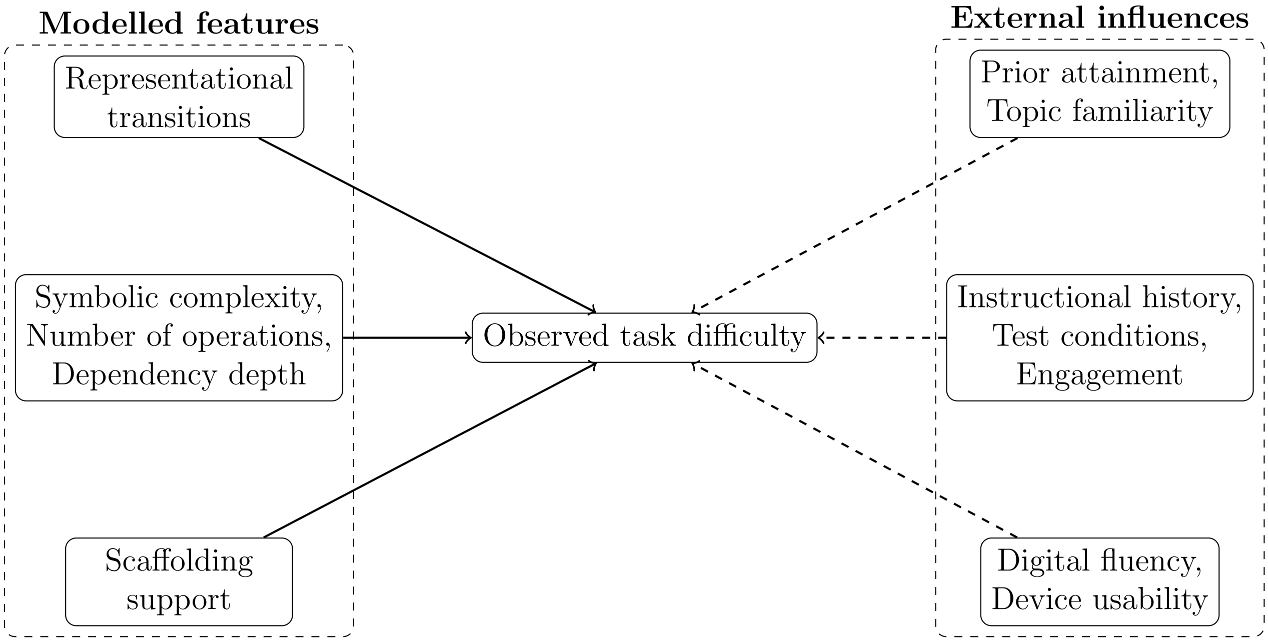


Figure 3.1: Conceptual factors influencing observed task difficulty. The proposed model explicitly captures structural and design features of tasks (left), while other external influences (right) are not directly modelled but may affect observed performance.

where θ_p represents the mathematical ability of student p, and b_i represents the difficulty of the algebra task i. The function σ is based on the logistic function

$$\sigma(x) = \frac{1}{1 + e^{-x}},$$

which maps real numbers to the interval $(0, 1)$ and is therefore suitable for modelling probabilities.

Expanding the logistic function for $x = \theta_p - b_i$ gives

$$\sigma(\theta_p - b_i) = P(X_{pi} = 1) = \frac{1}{1 + \exp\big(-(\theta_p - b_i)\big)},$$

Student ability is modelled as a latent variable θ, representing underlying proficiency that cannot be directly measured but is inferred from response patterns.

A key feature of the Rasch model is that the probability of a successful response depends only on the difference $\theta_p - b_i$. In particular, when $\theta_p = b_i$

$$\Rightarrow \; P(X_{pi} = 1) = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5,$$

so task difficulty can be interpreted as the ability level at which a student has a 50% chance of answering correctly. For example, if a task has difficulty 1.2, a student with ability level 1.2 has a 50% chance of answering correctly.

Why task difficulty should be explained

In the standard Rasch model, each task has a single difficulty parameter b_i, which is estimated from response data. While this provides a numerical measure of how difficult a task is, it does not explain why the task is difficult.

For algebraic tasks, difficulty may arise from identifiable features such as symbolic complexity, number of operations, or the need to coordinate multiple representations. The aim of this study is therefore to model b_i as a function of observable features of the task, rather than treating it as an unexplained parameter.

Observable features of algebra tasks

To explain task difficulty, we define a set of measurable features for each algebra task i. These are observable features of the task, including both structural and design-related components, and we assume that difficulty is influenced by these. Methods for measuring these features are set out in Appendix A.

Structural features:

These relate to the symbolic structure of the task:

- s_1: symbolic variety, being the number of distinct symbolic or numerical elements that must be tracked. Repeated occurrences of the same symbol are counted once. For example, in $3x + x + 2 = 11$ the distinct tokens are $\{x, 3, 2, 11\}$, so $s_1 = 4$. This variable captures symbolic variety rather than frequency.

- s_2: the optimal number of algebraic operational steps required for solution. Operations include substitution steps, addition and subtraction, multiplication and division, exponentiation, etc. The count refers to the steps needed to solve the problem, not simply the number of symbols in the question. For example, solving $3x + 2 = 11$ requires subtracting 2 and dividing by 3, so $s_2 = 2$. This variable captures procedural difficulty.

- s_3: degree of nesting or dependency within the expression. This captures how deeply operations are embedded within one another. Tasks involving nested brackets, powers, or substitution require students to coordinate multiple intermediate steps and are therefore expected to impose greater cognitive demand (Sweller, 1988). For example, a simple linear equation $3x+2 = 11$ has depth 1. A nested expression such as $2(x + 3)^2 - 5 = 7$ has depth 3 due to exponential function and inner grouping.

Representational and design features

These variables capture aspects of task structure and presentation that may influence effective cognitive demand and therefore task difficulty.

- d_1: number of required transitions between representations. For example, symbol $\leftrightarrow$ graph $\leftrightarrow$ words).

- d_2: count of supportive scaffolding features present in the task. These include: (i) breaking the task into steps, (ii) making intermediate structure visible, (iii) linking representations meaningfully, and (iv) prompting an appropriate method without removing the underlying mathematics.

Representational transition (d_1) is expected to increase difficulty, as tasks requiring multiple representations to be processed or coordinated simultaneously impose greater cognitive demand. Scaffolding (d_2) is usually expected to reduce effective difficulty by aiding coordination and structuring the task, although if poorly designed it can introduce additional processing requirements. The net effect of task design therefore depends on the balance between these components.

A structured model for task difficulty

We now decompose task difficulty into two components: structural difficulty and design-related difficulty. This reflects the distinction between the inherent mathematical demand of a task and the way in which it is presented to the learner.

We define

$$b_i = \beta_0 + b_i^{(\text{struct})} + b_i^{(\text{design})}$$

where β_0 is the base-line difficulty, $b_i^{(\text{struct})}$ is the structural difficulty and $b_i^{(\text{design})}$ is the design-related difficulty. This decomposition separates the inherent mathematical demand of the task from effects arising from its presentation.

Structural difficulty $b_i^{(\text{struct})}$

Structural difficulty represents the inherent mathematical demand of the task, arising from its symbolic and representational structure. This is modelled as

$$b_i^{(\text{struct})} = \beta_1 s_1 + \beta_2 s_2 + \beta_3 s_3$$

These components capture symbolic complexity, procedural demand, dependency structure, and representational coordination. They are expected to increase cognitive demand and therefore increase task difficulty.

Design-related difficulty $b_i^{(\text{design})}$

Design-related difficulty represents the effect of presentation and instructional features on the effective demand of the task. This is modelled as

$$b_i^{(\text{design})} = \beta_4 d_1 + \beta_5 d_2.$$

Combined model

Combining these components gives

$$b_i = \beta_0 + \beta_1 s_1 + \beta_2 s_2 + \beta_3 s_3 + \beta_4 d_1 + \beta_5 d_2$$

This formulation follows the LLTM (Fischer, 1973), distinguishing between the underlying mathematical structure of a task and design or presentational features, which may alter effective difficulty without changing the required mathematical operations. The different coefficients are:

- β_0 is a baseline difficulty;
- $\beta_1, \beta_2, \beta_3$ measure the contribution of structural complexity effects s_1, s_2, s_3;
- β_4 measures representational coordination effect d_1;

- β_5 measures the effect of supportive scaffolding d_2;

The coefficients β_j determine how each feature contributes to task difficulty. Structural and representational coefficients are expected to be positive, reflecting increased cognitive demand. Supportive scaffolding is expected to reduce effective difficulty, while instructional overhead is expected to increase it, although the net effect of design depends on the balance between these components.

Structured item response model for algebra tasks

Substituting this expression for b_i into the Rasch model and writing in compact form gives

$$P(X_{pi} = 1) = \frac{1}{1 + \exp\Big(- \big(\theta_p - \beta_0 - \sum_j \beta_j z_{ij}\big)\Big)}. \tag{3.1}$$

where

- β_0 is baseline difficulty,
- β_j measures the contribution of feature j,
- z_{ij} is the value of feature j for task i.
- θ_p is the ability of pupil, p.

This formulation makes explicit that item difficulty is decomposed into observable structural and design-related task features within the response model, and allows:

- comparison of tasks based on their structural properties,
- analysis of how different features contribute to difficulty,
- investigation of how task design, including supportive scaffolding and instructional overhead, alters effective task demand.

The probability of success now depends not only on student ability but also on both the structural properties of the task and its design-related features, which may either increase or decrease effective difficulty.

4 Illustrative example

To demonstrate how the structured model operates in practice (addressing RQ3), we consider three algebra tasks of increasing complexity, including one example with digital scaffolding. The aim of this section is not to estimate parameters from data, but to show explicitly how the proposed model converts structural and design features into task difficulties and predicted response probabilities.

4.1 Example tasks

We consider the following three tasks:

1. **Linear task (baseline):** Solve $11x + 7 = 11 + 7x$.
2. **Quadratic task (manual):** Solve $x^2 + 2x - 3 = 0$.

3. **Quadratic task (scaffolded):** Solve $x^2 + 2x - 3 = 0$, using digital support, as shown in Figure 1.1.

The first task is structurally simpler than the quadratic tasks. The second and third tasks have the same underlying algebraic structure. A graphical representation in the scaffolded task is offered and optional scaffolding (a slider and a hover-over).

Feature specification

Task	s_1	s_2	s_3	d_1	d_2
Linear	3	3	1	0	0
Quadratic (manual)	5	4	2	0	0
Quadratic (scaffolded)	5	4	2	1	-2

Here:

- s_1 denotes the number of symbolic elements,
- s_2 denotes the number of algebraic operations required,
- s_3 denotes dependency depth,
- d_1 denotes the number of representational transitions,
- d_2 denotes the number of supportive scaffolding features,

Parameter specification

For illustration purposes, we use the following representative coefficient values:

$$\beta_0 = 0.20, \quad \beta_1 = 0.10, \quad \beta_2 = 0.15, \quad \beta_3 = 0.30, \quad \beta_4 = 0.25, \quad \beta_5 = 0.40,$$

The larger the coefficient the more the impact on task difficulty.

4.2 Computed task difficulties

The structured difficulty model is

$$b_i = \beta_0 + \beta_1 s_1 + \beta_2 s_2 + \beta_3 s_3 + \beta_4 d_1 + \beta_5 d_2$$

Applying this to the three illustrative tasks gives:

Linear task

$$\begin{aligned} b_1 &= 0.2 + 0.1(3) + 0.15(3) + 0.3(1) + 0.25(0) + (0.4)(0) \\ &= 0.2 + 0.3 + 0.45 + 0.3 = 1.25. \end{aligned}$$

Quadratic task (unscaffolded)

$$\begin{aligned} b_2 &= 0.2 + 0.1(5) + 0.15(4) + 0.3(2) + 0.25(0) + (0.4)(0) \\ &= 0.2 + 0.5 + 0.6 + 0.6 = 1.9. \end{aligned}$$

Quadratic task (scaffolded)

$$\begin{aligned} b_3 &= 0.2 + 0.1(5) + 0.15(4) + 0.3(2) + 0.25(1) + (0.4)(-2) \\ &= 0.2 + 0.5 + 0.6 + 0.6 + 0.25 - 0.8 = 1.35 \end{aligned}$$

Interpretation of task difficulties

The resulting ordering is: $b_1 < b_3 < b_2$. This would indicate that:

- the linear task is the easiest,
- the unscaffolded quadratic task is the most difficult,
- the scaffolded quadratic task remains harder than the linear task, but is easier than the equivalent unscaffolded quadratic task.

The difference between b_2 and b_3 is due entirely to design features rather than algebraic structure. Since the two quadratic tasks have identical symbolic and operational structure, the reduction in difficulty from 1.90 to 1.35 is attributable to the net effect of scaffolding and the additional representational transition in the digital version. This demonstrates how the model can distinguish between symbolic complexity and instructional support.

4.3 Response probabilities for selected ability levels

To illustrate how task difficulty interacts with student ability, we compute predicted probabilities of success for three representative ability levels: $\theta = 0.50$, $\theta = 1.35$, $\theta = 2.0$. Using the Rasch model,

$$P(X_{pi} = 1) = \frac{1}{1 + \exp\big(-(\theta_p - b_i)\big)},$$

together with the illustrative difficulties: $b_1 = 1.25$, $b_2 = 1.9$, $b_3 = 1.35$ gives the following probabilities of success (to 2 d.p).

Task	$\theta = 0.5$	$\theta = 1.35$	$\theta = 2.0$
Linear ($b_1 = 1.25$)	0.32	0.52	0.68
Quadratic (unscaffolded, $b_2 = 1.90$)	0.20	0.37	0.52
Quadratic (scaffolded, $b_3 = 1.35$)	0.30	0.50	0.66

Table 4.1: Predicted probabilities of success for the three illustrative tasks at selected ability levels.

These values illustrate several important features of the model:

- When $\theta = b$, the probability of success is 0.5. This is observed for the scaffolded quadratic task at $\theta = 1.35$, confirming that its difficulty corresponds to the ability level required for a 50% chance of success.
- The scaffolded quadratic task has a higher probability of success than the equivalent unscaffolded quadratic task at each ability level.
- The difference between the probability of success for the two quadratic tasks is narrower for students whose ability is further from the level of task difficulty; scaffolding has less impact on those students.
- Higher-ability students have relatively high probabilities of success on all three tasks, while lower-ability students have low probabilities of success, although these are slightly higher for the scaffolded task.

This shows how differences in task difficulty translate into differences in predicted performance across the ability scale.

4.4 Graphical illustration

The relationship between ability and success probability for the three illustrative tasks is shown in Figure 4.1. The linear task has the lowest difficulty and therefore the leftmost response curve. The unscaffolded quadratic task is the most difficult, while the scaffolded quadratic task lies slightly to its left. This means that, for any fixed probability of success, a lower level of ability is required for the scaffolded task than for the equivalent manual task.

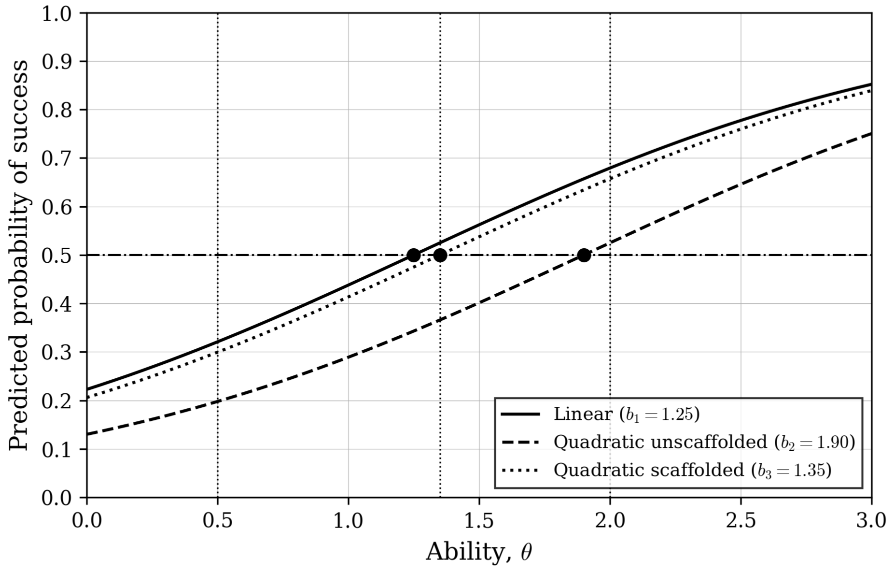


Figure 4.1: Predicted response curves for the three illustrative tasks. The dashed vertical lines show the selected ability values $\theta = 0.50$, $\theta = 1.35$, and $\theta = 2.00$ used in Table 4.1.

In particular, each curve crosses the line $P(X = 1) = 0.5$ at $\theta = b$, so task difficulty can be interpreted as the ability level required for a 50% probability of success. A reduction in difficulty therefore corresponds to a leftward shift of the response curve.

4.5 Role of the illustrative example

This illustrative example is intended to clarify how the model behaves, but does not account for variability in student responses. For this reason, the illustrative example acts as a conceptual demonstration rather than an empirical test. To assess whether such structural effects can be reliably detected when responses are subject to randomness, a simulation study is required.

5 Simulation design

5.1 Purpose and context of the simulation

The illustrative example in the previous section demonstrates how the proposed model converts structural and design features into task difficulties and response probabilities. However, it is deterministic: once the coefficients and feature values are specified, the resulting difficulties follow directly. In practice, student responses are random. A simulation study is therefore needed.

The purpose of the simulation study is not to establish that the proposed model is true, but to examine whether the estimation framework is capable of recovering a known structural effect (known parameter values) under realistic, noisy response variation. This follows the general logic of simulation studies in Rasch measurement e.g. De Boeck and Wilson (2004), where responses are generated under known parameter values and the extent to which those parameters can be recovered is then evaluated.

The simulation examines whether a known reduction in difficulty associated with scaffolding can be detected from generated response data, evaluating how accurately the original parameters $\beta_0, \beta_1, \ldots, \beta_5$ can be recovered using logistic regression.

5.2 Model specification

From Chapter 3, the probability that student p correctly answers item i is given by:

$$P(X_{pi} = 1) = \frac{1}{1 + \exp\Big(- \big(\theta_p - b_i\big)\Big)} \tag{5.1}$$

$$\text{where} \quad b_i = \beta_0 + \beta_1 s_1 + \beta_2 s_2 + \beta_3 s_3 + \beta_4 d_1 + \beta_5 d_2. \tag{5.2}$$

Recall that the parameters are

- θ_p represents the latent ability of student p,
- b_i represents the difficulty of item i,
- β_0 is baseline difficulty,
- β_j measures the contribution of feature j,
- z_{ij} is the value of feature j for task i,
- $X_{pi} \in \{0, 1\}$ is the simulated response,

5.3 Simulation inputs and design choices

Fixed values for the parameters $\beta_0, \beta_1, \ldots, \beta_5$ are specified to define the data-generating process for 40 example questions. For each of the 40 example questions, values of the predictors s_1, s_2, s_3, d_1, and d_2 are calculated, defining item characteristics.

The number of students n is chosen to determine the sample size for the simulation. Larger values of n are expected to improve the accuracy of parameter recovery.

Parameter generation

Synthetic data were generated to reflect a realistic assessment scenario.

Student abilities θ_i are simulated from a normal distribution for the required number of students:

$$\theta_i \sim \mathcal{N}(3, 0.5^2), \tag{5.3}$$

representing a realistic spread of attainment at lower secondary school KS3 level. Simulations were ran for $n = 500$ students.

Item features were specified deterministically based on task design principles contained in Appendix A (for example, algebraic complexity or number of steps), and corresponding feature effects s_1, s_2, s_3, d_1 and d_2 were calculated to produce a range of item difficulties for 40 tasks.

Reasonable β_0-β_5 parameters were included, with the purpose of the simulation being to recover these parameters.

Pupil response simulation

For each student–question pair, the probability of a correct response p_{ij} is computed using the logistic model. These probabilities are stored in an $n \times 40$ matrix. Binary responses were then generated using the Bernoulli distribution:

$$X_{ni} \sim \text{Bernoulli}(P(X_{ni} = 1)) \tag{5.4}$$

This produced a response matrix to indicate whether the pupils gave correct (1) or incorrect (0) responses to each question, consistent with a Rasch-style probabilistic framework. Figure 5.1 illustrates individual student outcomes for each task, which is then shown as the full response matrix (500x40) in figure 5.2 as a black and white heat map.

		Task difficulty					
		0.95	1.1	0.9	...	4.85	5
Student ability	1.59	0	0	0	...	0	0
	1.75	0	1	1	...	0	0
	1.79	0	1	1	...	0	0
	⋮	⋮	⋮	⋮	⋮	⋮	⋮
	4.27	1	1	1	...	0	1
	4.63	1	1	1	...	0	0

Figure 5.1: Pupil response data

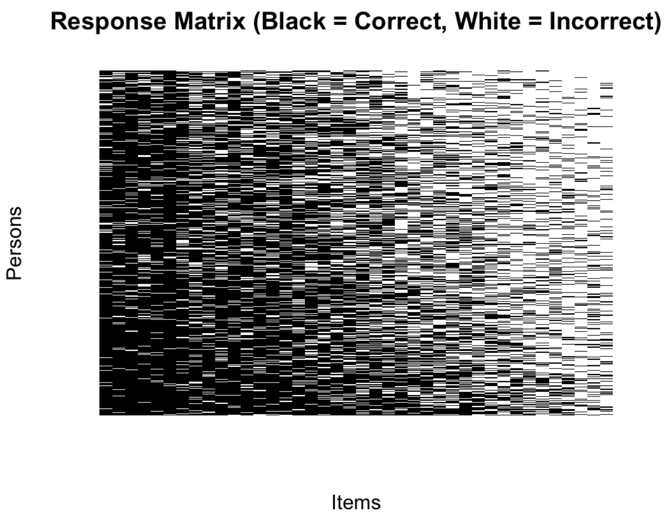


Figure 5.2: Response matrix

5.4 Parameter recovery

Data restructuring

The simulated response matrix was reshaped into long format for regression analysis. Each observation contains: student identifier, question identifier, response (0 or 1), and predictors s_1, s_2, s_3, d_1, d_2.

Parameter recovery

A logistic regression model is fitted using a Generalised Linear Model (GLM) with logit link and Bernoulli likelihood to determine the parameter values. In this way it attempted to recover student abilities θ_n and feature effects β_1 to β_5.

Given the model formulation:

$$\text{logit}(P(X_{pi} = 1)) = \theta_p - b_i, \tag{5.5}$$

where

$$\text{logit}(P) = ln\left(\frac{P}{1-P}\right) \tag{5.6}$$

the estimated regression coefficients correspond to $-\beta$. Therefore, sign adjustment is required to recover the original parameters.

Evaluation of parameter recovery

Model performance was evaluated by comparing estimated parameters with their true simulated values, considering the error.

Root Mean Square Error (RMSE) was used to quantify recovery accuracy:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{\beta}_i - \beta_i)^2} \tag{5.7}$$

Comparisons were made between estimated ($\hat{\theta}$) and true (θ) abilities, and true and estimated item difficulty parameters.

Graphical analysis included scatter plots of true versus estimated parameters to assess alignment between ability and difficulty distributions.

5.5 Model interpretation and limitations

While the model retains the Rasch structure on a logit scale, the inclusion of a baseline term and explanatory features allows task difficulty to be interpreted in terms of observable characteristics. The estimated coefficients indicate how each feature contributes to item difficulty, with positive values increasing difficulty and negative values reducing it.

Student ability is simulated from a normal distribution and assumed constant across items for each individual. Although ability is theoretically a latent variable (not directly observed but inferred from response patterns), in this model it is approximated using person-specific fixed effects.

However, the model remains only partially explanatory. Any unobserved influences on performance, such as prior knowledge or strategy use, are absorbed into the error term and are not explicitly modelled. In addition, the assumption that task difficulty is a linear combination of features may oversimplify the relationship between task structure and cognitive demand.

5.6 Assumptions

The simulation assumes:

- normally distributed student ability,
- a logistic relationship between ability and response probability,
- conditional independence of responses,
- item difficulty fully explained by the chosen predictors.

6 Simulation results

6.1 Parameter recovery accuracy

Table 6.1 presents the true parameter values alongside the estimated values obtained from the logistic regression model compared to true values. Overall, the model demonstrates good parameter recovery, with all coefficients estimated within approximately 0.01 of their true values.

To assess parameter recovery more formally, the confidence intervals in Figure 6.2 can be interpreted using a hypothesis testing framework. For each parameter, the null hypothesis is that the estimated coefficient is equal to the true value used in the simulation:

$$H_0 : \beta_j = \beta_j^{\text{true}},$$

against the two-sided alternative

$$H_1 : \beta_j \neq \beta_j^{\text{true}}.$$

Using a 95% confidence interval is equivalent to carrying out a two-sided hypothesis test at the 5% significance level, with 2.5% in each tail. Since the true parameter values (indicated by the red cross) lie within the 95% confidence intervals for all estimated coefficients, we fail to reject H_0 for each β_j. This indicates that there is no statistical evidence, under the simulated conditions, that the recovered coefficients differ significantly from the true values used to generate the data. It also provides further support for the accuracy of the parameter recovery process. In particular, it suggests that the estimation method is able to recover the underlying feature effects reliably.

Parameter	True Value	Estimate	Error
β_1	0.100	0.110	0.010
β_2	0.150	0.150	0.000
β_3	0.300	0.300	0.000
β_4	0.250	0.239	-0.011
β_5	0.400	0.413	0.013

Figure 6.1: Parameter recovery results (to 3 decimal places)

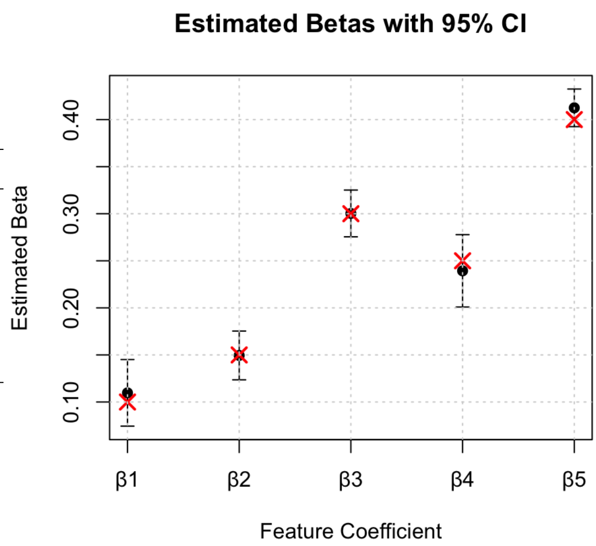


Figure 6.2: Confidence intervals for estimated parameters

6.2 Effect of sample size

To investigate the effect of sample size, the simulation was repeated for $n = 20, 50, 200, 500$. Figure 6.3 shows that RMSE decreases exponentially as sample size n increases. This behaviour is consistent with asymptotic properties of maximum likelihood estimators, where variance decreases with larger sample sizes. Larger samples provide more information about the likelihood function, leading to more stable and precise parameter estimates. The improvement is particularly pronounced between $n = 50$ and $n = 200$, after which gains become more gradual.

This suggests that, in applied settings, reliable estimation of item parameters requires sufficiently large datasets, particularly when multiple predictors are included. Sample sizes of 200, which would be two year groups in a large secondary school, would provide sufficient datasets.

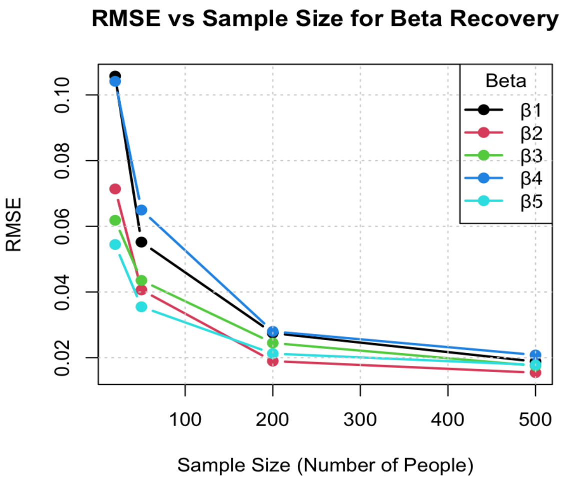


Figure 6.3: RMSE as a function of sample size for β parameters.

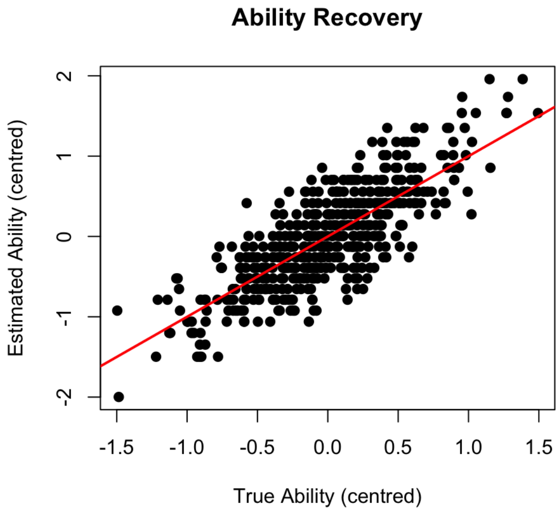


Figure 6.4: True vs estimated ability.

6.3 Model fit: true vs estimated abilities

Figure 6.4 compares true and estimated student abilities. There is a clear positive linear relationship, with most points lying close to a 45-degree line.

To quantify this relationship, the Pearson correlation between true and estimated abilities was calculated as 0.8037, indicating a strong positive association. This suggests that the model is able to recover relative differences in ability effectively, although some deviation is observed, particularly at the extremes of the distribution.

This is important because accurate estimation of student ability allows the model to separate person and item effects, ensuring that the estimated feature coefficients reflect task difficulty rather than differences in student performance. This supports the validity of the parameter recovery results.

6.4 Interpretation of parameter recovery

The confidence interval and RMSE results provide further support for the effectiveness of parameter recovery. The confidence intervals indicate that the estimated coefficients are consistent with the true values, while the RMSE trend shows that estimation error decreases with increasing sample size and stabilises for larger n.

Taken together, these results suggest that the model is able to recover the underlying parameters accurately and with increasing precision, supporting the validity of the simulation approach.

6.5 Evaluation

The simulation demonstrates that logistic regression can recover item difficulty parameters with a high degree of accuracy under controlled conditions. Increasing sample size improves estimation precision, while the inclusion of interpretable predictors allows insight into the drivers of item difficulty.

However, comparison with the Rasch model highlights important conceptual differences, particularly regarding the treatment of latent ability. This suggests that while GLMs provide a useful approximation, full IRT models may be required for more robust inference in real-world educational data.

6.6 Interpretation of results

The estimated coefficients provide insight into the relative importance of different features in determining task difficulty. Scaffolding has the largest effect in magnitude ($\beta_5 = 0.413$), indicating that design-related features can substantially alter the effective difficulty of a task. Importantly, this effect may either increase or decrease difficulty depending on how scaffolding is implemented.

Among the structural features, dependency depth has the strongest effect ($\beta_3 = 0.300$), suggesting that tasks with more nested or interdependent steps are significantly more demanding. Representational transitions also contribute meaningfully to difficulty ($\beta_4 = 0.239$), reflecting the additional cognitive effort required to coordinate multiple forms. In comparison, the number of algebraic operations ($\beta_2 = 0.150$) and symbolic elements ($\beta_1 = 0.110$) have smaller effects.

Overall, these results suggest that deeper structural complexity and task design features have a greater impact on difficulty than surface-level increases in symbols or procedural steps.

7 Limitations of the simulation

7.1 Limitations

The simulation is conducted under a highly idealised data-generating process, in which responses are generated using the same model that is later used for estimation. This creates favourable conditions for parameter recovery, as the model is correctly specified by construction. In practice, real student response data are unlikely to follow the model exactly, meaning that the accuracy observed in the simulation may overstate the performance of the estimation procedure in applied settings. In addition, the simulation assumes that responses depend only on student ability and task difficulty, omitting other sources of variability such as guessing, careless errors, fatigue, and differences in student engagement. These factors introduce additional noise in real-world data and may reduce the precision of parameter estimates.

Further simplifications arise in the treatment of student ability and task features. Ability is assumed to follow a normal distribution, which may not reflect the structure of real classroom populations, where attainment can be skewed or clustered. Task features are also treated as fixed and perfectly measured, whereas in practice their coding involves a degree of subjectivity and may be prone to measurement error. Moreover, the simulation assumes that the effects of task features are purely additive, meaning that each feature contributes independently to task difficulty. In reality, the impact of one aspect of a task may depend on the presence of others, and such relationships are not captured within the current framework. Finally, the simulated tasks are defined through predetermined feature values rather than drawn from authentic classroom materials, and all students are assumed to respond to every item. This limits the realism of the simulation, as real datasets often contain more varied task structures and incomplete response patterns, for example due to time constraints.

7.2 Extensions

These limitations suggest several opportunities for extending the simulation to better reflect real-world conditions. One important extension would be to introduce additional sources of variability into the data-generating process, such as random response errors or guessing behaviour, in order to assess how robust parameter recovery remains under less idealised conditions. Similarly, alternative distributions of student ability could be explored, including skewed or mixed distributions, to better represent realistic classroom populations. The simulation could also be extended to incorporate uncertainty in the coding of task features, allowing for the investigation of how measurement error in variables such as symbolic complexity or scaffolding influences the stability of estimated parameters.

Further extensions could aim to capture more complex relationships between task characteristics and performance. For example, the effect of scaffolding could be allowed to vary across different levels of task complexity or student attainment, providing a more nuanced representation of how digital support operates in practice. In addition, using task sets derived from authentic classroom materials, rather than predefined feature combinations, would improve the pedagogical validity of the simulation. Finally, introducing incomplete response patterns or missing data would allow the robustness of the model to be evaluated under more realistic testing conditions. Together, these extensions would strengthen the connection between the simulation and real educational settings, providing a more comprehensive assessment of the model's practical applicability.

8 Empirical application of the model

The main value of the model lies in its application to real student response data, where it provides a structured framework for comparing algebra task performance and testing hypotheses about the effects of different task presentations. In particular, this addresses the central research question (RQ4) of how digital presentation may influence algebra performance. By expressing task difficulty as a function of observable structural and design features, the model allows the effects of digital representations and scaffolding to be isolated and quantified on a common logit scale, enabling differences in performance to be interpreted in a statistically meaningful way.

The framework can be used to compare algebra tasks presented in digital and non-digital formats by holding structural features constant and isolating the contribution of design-related variables such as representational transitions and scaffolding. This enables differences in performance to be attributed to presentation effects rather than underlying mathematical complexity. On this basis, a set of testable hypotheses can be formulated regarding the determinants of task difficulty within the model.

- **H1 (Effect of scaffolding):** Digitally scaffolded tasks are expected to have lower effective difficulty than equivalent non-scaffolded versions of the same task.
- **H2 (Prior knowledge and digital support):** The impact of digital presentation on algebra task performance is expected to depend on students' prior knowledge, with supportive scaffolding features reducing effective difficulty more for lower-attaining students, while offering limited benefit to higher-attaining students.
- **H3 (Overall effect of digital presentation):** Digital task presentation may lead to improved performance compared to traditional formats when supportive features outweigh any additional cognitive demands, after controlling for relevant confounding factors.

To evaluate these hypotheses, an analytical framework is required that allows the contribution of structural and design-related features to task difficulty to be estimated from response data. This can be achieved using a Rasch-style logistic regression model, in which the probability of a correct response depends on the difference between student ability and task difficulty. Within this framework, task difficulty is expressed as a function of observable features, enabling the effects of variables such as representational transitions and scaffolding to be estimated through regression coefficients. The hypotheses can then be tested by examining the sign, magnitude, and statistical significance of the estimated coefficients associated with key variables. In particular, the effect of scaffolding (H1) can be assessed through the coefficient on supportive features, while the overall impact of digital presentation (H3) can be examined by comparing tasks with similar structural characteristics but differing design features. The hypothesis relating to prior knowledge (H2) requires a slightly different approach and can be investigated by comparing the effects of digital task features across groups of students with differing levels of prior attainment, or by extending the model to allow for interactions between ability and design-related variables. In this way, the framework provides a coherent method for evaluating how digital task design influences algebra performance while controlling for both task structure and variation in student ability.

To apply this framework in practice, a suitable dataset would consist of student responses to a range of KS3 algebra tasks presented in both digital and non-digital formats. These responses would be recorded in binary form (correct or incorrect) and linked to both student and task identifiers. The task set should include items with varying levels of structural complexity, alongside matched examples that share the same underlying algebraic structure but differ in presentation, allowing for direct comparison between formats. For each task, values of the structural and design-related features $(s_1, s_2, s_3, d_1, d_2)$ would be systematically coded according to a consistent scheme, such as that outlined in Appendix A. In addition, information on students' prior attainment would be collected, for example through previous assessment scores, enabling the analysis of differences in performance across attainment levels. The resulting dataset would be organised in long format, with each observation corresponding to a student–task interaction, and including both response outcomes and associated predictors. Where possible, tasks would be administered to the same group of students, or allocated in a controlled manner, to ensure comparability across formats and reduce potential sources of bias. This structure provides the necessary inputs for estimating the model and examining how variation in task design and student characteristics relates to observed performance.

In applying this framework, careful consideration must be given to potential confounding factors and the overall study design. A key strength of the Rasch-based approach is that it explicitly accounts for variation in student ability, allowing differences in performance to be interpreted independently of underlying attainment. However, other factors may influence results, including prior knowledge of specific topics, familiarity with digital tools, and differences in classroom conditions or instructional context. To mitigate these effects, tasks should be designed to be comparable in content and difficulty, with matched pairs of digital and non-digital items where possible, ensuring that differences in performance can be attributed to presentation rather than underlying structure. Ideally, the same group of students would complete tasks in both formats, or tasks would be allocated in a controlled or randomised manner to reduce selection bias. Additional variables, such as prior attainment, can be incorporated into the analysis to further account for differences between learners. Potential sources of bias also include variation in student engagement, differences in digital fluency, and subjectivity in the coding of task features. While these factors cannot be fully eliminated, careful task design, consistent coding procedures, and appropriate controls help to minimise their impact. Overall, these considerations strengthen the validity of the analysis by increasing confidence that observed differences in performance reflect the effects of task design rather than external influences.

Based on the proposed framework, it is expected that structural features will be positively associated with task difficulty, while supportive scaffolding will reduce effective difficulty, particularly for lower-attaining students. Digital presentation is therefore unlikely to have a uniform effect, instead altering task difficulty through a balance between increased representational demand and instructional support.

This empirical testing of the model may suggest areas for refinement, including the incorporation of interaction effects between task features and student characteristics, as well as improvements in the consistency and objectivity of feature coding. Such developments would enhance the model's ability to capture the complexity of real classroom settings.

In practical terms, the model provides a structured approach for evaluating and designing algebra tasks, particularly within digital learning environments. It offers a means of predicting how changes in presentation may influence performance before implementation, supporting more informed instructional decisions. However, ethical considerations must also be taken into account. In particular, the use of such models should not lead to oversimplified assumptions about learner ability or restrict access to more challenging tasks for certain groups of students. Instead, the framework should be used to support equitable access to appropriately challenging and well-designed learning experiences.

9 Conclusion

This project set out to develop a structured model of algebra task difficulty and to examine how such a framework could be used to analyse the effects of digital presentation on student performance. The aim was to move beyond treating difficulty as a single latent parameter and instead provide a framework that links task difficulty to observable features of algebra tasks.

To achieve this, a model was constructed combining ideas from item response theory, cognitive load theory, and research on mathematical representations. Task difficulty was expressed as a function of structural and design-related features, allowing both the inherent mathematical complexity of a task and the effects of its presentation to be captured within a single framework. A simulation study was then conducted to assess whether these feature effects could be reliably recovered from response data.

The simulation results demonstrated that, under controlled conditions, the model is capable of recovering the underlying parameters with a high degree of accuracy. In particular, the findings show that task difficulty can be decomposed into interpretable components, allowing the contribution of symbolic complexity, dependency structure, representational transitions, and scaffolding to be quantified. The analysis also highlights that digital presentation does not have a uniform effect on performance, but instead alters task difficulty through a balance between increased cognitive demand and supportive design features.

While the analysis provides useful insights, it is based on simplified assumptions and idealised data. Real student responses are likely to involve additional sources of variability, and further empirical work is required to evaluate how well the framework performs in classroom settings and under more complex conditions.

Overall, the project contributes a structured and mathematically grounded approach to analysing algebra task difficulty in relation to digital presentation. By linking difficulty to observable task features, the model provides a tool for evaluating and designing algebra tasks prior to implementation. This has practical implications for curriculum design and digital learning environments, supporting a more systematic and evidence-based approach to task design in mathematics education.

A Coding of task features

Each feature should be coded in a consistent and reproducible way based on observable characteristics of the task. This appendix defines how task features can be consistently identified for use in the structured item response model.

First read the task and any supports. Apply each measure independently. Enter scores as item predictors in Rasch analysis.

Structural features

- s_1: symbolic variety. This is the number of distinct symbolic or numerical elements that must be tracked in the task. Repeated occurrences of the same symbol are counted once. (e.g. in $3x + x + 2 = 11$, the distinct elements are $\{x, 3, 2, 11\}$, so $s_1 = 4$)

- s_2: number of algebraic operations required for solution. This refers to the minimum number of procedural steps needed to solve the task.(e.g. $3x + 2 = 11$ requires subtracting 2 and dividing by 3, so $s_2 = 2$)

- s_3: degree of nesting or dependency. This reflects the depth of embedded operations within the expression, which could be independent (1), one step supporting another (2), sequential steps (3), or highly integrated (4).(e.g. $3x + 2 = 11$ has depth 1, while $2(x + 3)^2 - 5 = 7$ has depth 3 due to nested brackets and exponentiation)

Design features

This encompasses the representational format of the question, and any scaffolding.

- d_1: number of representations to transition between (e.g. symbolic to graphical or verbal). Count the distinct formats used: written, physical, visual/diagrammatic (bar, number line, graph), symbolic, verbal, tabular. (e.g. solving an equation purely algebraically gives $d_i = 1$, while solving using a graph as well as an equation gives $d_1 = 2$)

- d_2: number of scaffolding features, which can be beneficial or detrimental. Each feature is coded as present or absent according to the table below, and the total d_2 is obtained by summing across all features. The range is roughly 0–12; higher = more support, meaning expected lower difficulty.

Scaffolding may be:

- Conceptual: from basic (e.g., "think of a balance") ranging to deep (multiple linked models + "why it works").

- Procedural: simple full worked example; partial steps/prompts; faded checklist (student completes increasing portions).

- Metacognitive: could be a basic check prompt ("substitute back"); to full planning/reflection ("plan your steps, check errors").

- Adaptive: may not be present (static/fixed for all); basic error-specific hints; to real-time/personalised (dynamic adjustment).

Table A.1: Scaffolding scoring for digital algebra tasks

Feature	Increases difficulty (+)	Neutral (0)	Reduces difficulty (-)	Typical range
Conceptual	Misleading or conflicting representations obscure mathematical structure.	No explicit conceptual support.	Cues; explicit relational links; interactive links.	$+1$ to -1
Procedural	Confusing, rigid or excessive steps/clicks increase extraneous load.	No procedural guidance.	Partial prompts; guided steps; adaptive clickable step-by-step support.	0 to -2
Meta-cognitive	Prompts encourage guessing, superficial checking or dependency.	No planning, monitoring or reflection support.	Basic checking; structured planning, monitoring and reflection.	0 to -1
Adaptivity	Poorly timed or distracting feedback disrupts problem-solving.	Static task with identical support for all learners.	Responsive hints; personalised feedback or difficulty adjustment.	$+1$ to 0

Example scaffolded digital task: Interactive app for solving $3x + 5 = 17$.

- Conceptual: Hover-over balance explanation, links to worked example → Score **-2**

- Procedural: Clickable step-by-step → Score **-1**

- Metacognitive: "Try again" button encourages guessing → Score **+1**

- Adaptivity: Hints adjust after two wrong attempts → Score **-1**

- **Total score = -3**

List of acronyms

GLM Generalised Linear Model. 19, 22

IRT Item Response Theory. 2, 7–9, 22

KS3 Key Stage 3. , 2, 9, 18, 25

LLTM Linear Logistic Test Model. 7–10, 13

RMSE Root Mean Square Error. 20–22

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